

Miguel Melo

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EDUCATION

University of South Florida
B.S., Computer Science

Expected Graduation: May 2026
GPA: 3.61/4.0

SKILLS

- **Languages:** C++, Python, C, HTML, CSS, Flask, JavaScript, Verilog, RISC-V
- **Technologies/Tools:** Flask, GitHub, Power Bi, CAD, Markdown, Bootstrap
- **Additional Skills:** Excel, Microsoft Azure, Copilot, DevOps
- **Interests:** Discrete Math, Logic Problems and Puzzles, Game Development

WORK EXPERIENCE

Amazon

May 2025 – August 2025

Software Developer Engineer Intern – AmazonAPI Team

Seattle, WA

- Upgraded internal detection systems for API contract privacy risks and contract violations, preventing exposure of sensitive fields before deployment into production
- Leveraged Large Language Models (LLMs) to automate API contract analysis, enabling early detection of missing privacy metadata, insecure fields, and formatting inconsistencies
- Engineered automated validation checks that flagged API contract violations, reducing manual compliance review workload by 40% for partnering teams
- Collaborated with senior SDEs in an Agile environment to design and deploy solutions aligned with Amazon's internal privacy and API governance standards
- Technologies Used: Python, AWS Cloud Services, CI/CD, API validation frameworks, internal Amazon tools

Eneva

May 2024 – August 2024

Intern – IT Architecture Team

Rio de Janeiro, Brazil

- Contributed to one of Brazil's top six gas-based power generation companies by enhancing cloud storage and API integration for optimized efficiency and performance
- Developed new APIs using Flask to facilitate secure communication with cloud databases hosted on Microsoft Azure, improving system accessibility
- Audited and refined 60+ team documents (ADRs and Playbooks), reducing errors by 50% and establishing a streamlined repository in GitHub for improved collaboration and version control
- Technologies Used: Power BI, Python, Flask, HTML, CSS, JavaScript, Markdown, Microsoft Azure

PROJECTS

AI Substitute Professor:

- Designed and implemented an LLM-driven AI teaching assistant sponsored by Microsoft, capable of delivering interactive lecture responses and answering student questions based on course-specific context
- Integrated Microsoft Azure services (Azure OpenAI, Blob Storage) to manage course content ingestion, retrieval, and real-time query handling with sub-second response times
- Developed and tested prompt-engineering and grounding strategies that reduced hallucinated responses and improved factual alignment with provided course materials during evaluations

String-Operation Converter:

- Developed a C++ program for a Data Structures course that reads, validates, and executes mathematical expressions, supporting a wide range of operators and decimal values
- Increased accuracy by incorporating robust input validation, resulting in 95% correct outputs across tests
- Employed linked lists and vectors for efficient memory management, handling expressions of up to 500 characters without performance issues